Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

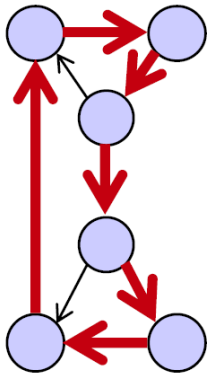
CS3230

Tutorial

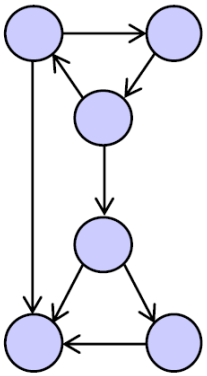
Week 13

Directed Hamiltonian Cycle

DIR-HAM-CYCLE: Given a digraph $G = (V, E)$, does there exists a simple directed cycle Γ that contains every node in V exactly once?



Has Hamiltonian cycle



No Hamiltonian cycle

Question 1

Give a poly time reduction from 3-SAT to DIR-HAM-CYCLE.